ICS 253 – Discrete Structures

Chapter 3: Sets

Question bank: 90 multiple choice + 10 written problems

Questions from ICS 253 old quizzes and homework (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

What is the cardinality of the set $\{1, 1, 2, 3, 3, 3\}$?

(A) 6

(B) 2

(C) 3

(D) 4

Answer: (C). Repeating an element does not change a set: $\{1, 1, 2, 3, 3, 3\} = \{1, 2, 3\}$, which has 3 distinct elements.

My answer

Q2. [new]

Which of the following statements is FALSE?

(A) $0 \in \mathbb{N}$

(B) $0 \in \mathbb{Z}^{+}$

(C) $-3 \in \mathbb{Z}^{-}$

(D) $\frac{1}{2} \in \mathbb{Q}$

Answer: (B). 0 is neither positive nor negative, so $0 \notin \mathbb{Z}^{+}$. In the zyBook $\mathbb{N}$ includes 0.

My answer

Q3. [new]

Which roster notation describes the set $\{x \in \mathbb{Z} : -2 < x \leq 3\}$?

(A) $\{-1, 0, 1, 2, 3\}$

(B) $\{-1, 0, 1, 2\}$

(C) $\{-2, -1, 0, 1, 2, 3\}$

(D) $\{-2, -1, 0, 1, 2\}$

Answer: (A). -2 is excluded (strict inequality) and 3 is included.

My answer

Q4. [new]

Which set-builder definition describes the set $\{1, 4, 9, 16, 25\}$?

(A) $\{x \in \mathbb{Z} : x \text{ is a perfect square and } x < 25\}$

(B) $\{x \in \mathbb{N} : x \text{ is a perfect square and } x \leq 25\}$

(C) $\{x \in \mathbb{Z}^+ : x \text{ is a perfect square and } x < 25\}$

(D) $\{x \in \mathbb{Z}^+ : x \text{ is a perfect square and } x \leq 25\}$

Answer: (D). (A) and (B) contain 0; (A) and (C) exclude 25.

My answer

Q5. [new]

What is $|\{x \in \mathbb{Z} : x \text{ is prime and } 10 \leq x \leq 30\}|$?

(A) 5

(B) 6

(C) 7

(D) 8

Answer: (B). The set is $\{11, 13, 17, 19, 23, 29\}$.

My answer

Q6. [new]

Which of the following sets is infinite?

(A) $\{x \in \mathbb{R} : 0 < x < 1\}$

(B) $\{x \in \mathbb{N} : x \text{ is a divisor of } 60\}$

(C) $\{x \in \mathbb{Z} : x^2 < 100\}$

(D) $\{x \in \mathbb{Z}^+ : x \text{ is prime and } x < 1000\}$

Answer: (A). There are infinitely many real numbers between 0 and 1; each of the other sets is a subset of a bounded set of integers.

My answer

Q7. [new]

Let $A = \{x \in \mathbb{R} : x^2 - 5x + 6 = 0\}$. Then $A =$

(A) $\{2, 3\}$

(B) $\{-2, -3\}$

(C) $\{1, 6\}$

(D) $\{-1, 6\}$

Answer: (A). $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$ gives $x = 2$ or $x = 3$.

My answer

Q8. [new]

Let $A = \{x \in \mathbb{Z} : x^2 \leq 4\}$, $B = \{-2, -1, 0, 1, 2\}$ and $C = \{x \in \mathbb{N} : x \leq 2\}$. Which statement is true?

(A) $A = C$

(B) $B \subseteq C$

(C) $A \subset B$

(D) $C \subset B$

Answer: (D). $A = B = \{-2, -1, 0, 1, 2\}$ and $C = \{0, 1, 2\}$. $C \subseteq B$ and $-1 \in B - C$, so $C \subset B$. $A \subset B$ fails because $A = B$.

My answer

Q9. [new]

Let $A = \{1, 2, 3, 4\}$, $B = \{2, 4\}$ and $C = \{2, 4, 5\}$. Which statement is true?

(A) $C \subseteq A$

(B) $A \subseteq B$

(C) $B \subset A$

(D) $C \subset B$

Answer: (C). Every element of B is in A and $3 \in A$ but $3 \notin B$. $C \not\subseteq A$ since $5 \notin A$.

My answer

Q10. [new]

Let $A = \{a, b, c\}$. Which statement is FALSE?

(A) $\emptyset \subseteq A$

(B) $A \subset A$

(C) $A \subseteq A$

(D) $\{a, c\} \subset A$

Answer: (B). A proper subset must miss some element of A; A is a subset of itself but not a proper subset.

My answer

Q11. [new]

Which of the following subset relationships between common number sets is FALSE?

(A) $\mathbb{Z} \subseteq \mathbb{Q}$

(B) $\mathbb{N} \subseteq \mathbb{Z}$

(C) $\mathbb{Z}^+ \subset \mathbb{N}$

(D) $\mathbb{Q} \subseteq \mathbb{Z}$

Answer: (D). $\frac{1}{2} \in \mathbb{Q}$ but $\frac{1}{2} \notin \mathbb{Z}$. (C) is true because $0 \in \mathbb{N}$ but $0 \notin \mathbb{Z}^+$.

My answer

Q12. [new]

Which statement is true for all sets A and B?

(A) If $A \subseteq B$, then $A \subset B$.

(B) If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

(C) If $x \in A$ and $A \subseteq B$, then $x \notin B$.

(D) If $A \subset B$, then $B \subseteq A$.

Answer: (B). Two sets are equal if and only if each is a subset of the other.

My answer

Q13. [new]

In a Venn diagram the universal set is $U = \{1, 2, 3, 4, 5\}$, the circle for A contains $1, 2, 3, 4$, and the oval for B lies completely inside the circle for A and contains only 2 and 4. Which relationship does the diagram show?

(A) $A \subseteq B$

(B) $A = B$

(C) $B \subset A$

(D) $B \not\subseteq A$

Answer: (C). $B = \{2, 4\} \subseteq A$ and $1, 3 \in A - B$, so B is a proper subset of A.

My answer

Q14. Quiz 2 Alenzi

Let $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{\{1, 2\}\}$. Which statement is true?

(A) $A \in B$

(B) $A \subseteq B$

(C) $\{A\} \subseteq \{B\}$

(D) $\{A\} \in B$

Answer: (A). B has exactly one element, the set $\{1, 2\} = A$. $A \not\subseteq B$ since $1 \notin B$; $\{A\} = \{\{1, 2\}\} \not\subseteq \{B\} = \{\{\{1, 2\}\}\}$; the only element of B is $\{1, 2\}$, not $\{\{1, 2\}\}$.

My answer

Q15. [new]

What is the cardinality of $A = \{1, \{1\}, \{1, 2\}, 3\}$?

(A) 3

(B) 4

(C) 5

(D) 6

Answer: (B). The elements are 1, $\{1\}$, $\{1, 2\}$ and 3; a set counts as one element.

My answer

Q16. [new]

Let $A = \{1, \{2, 3\}, \{4\}\}$. Which statement is true?

(A) $2 \in A$

(B) $\{2, 3\} \subseteq A$

(C) $4 \in A$

(D) $\{4\} \in A$

Answer: (D). The elements of A are 1, $\{2, 3\}$ and $\{4\}$. The numbers 2, 3, 4 are not elements of A.

My answer

Q17. [new]

Let $A = \{1, \{2, 3\}, \{4\}\}$. Which statement is true?

(A) $\{\{2,3\}\} \subseteq A$

(B) $\{2,3\} \subset A$

(C) $\{1,4\} \subseteq A$

(D) $\{\{4\}\} \in A$

Answer: (A). The only element of $\{\{2,3\}\}$ is $\{2,3\}$, which is an element of A.

My answer

Q18. [new]

What is $|\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}|$?

(A) 1

(B) 2

(C) 3

(D) 4

Answer: (C). The three elements $\emptyset$, $\{\emptyset\}$ and $\{\emptyset, \{\emptyset\}\}$ are distinct.

My answer

Q19. [new]

Which statement is FALSE?

(A) $\emptyset \in \{\emptyset\}$

(B) $\emptyset \subseteq \{\emptyset\}$

(C) $\{\emptyset\} \neq \emptyset$

(D) $|\{\emptyset\}| = 0$

Answer: (D). $\{\emptyset\}$ has exactly one element (the empty set), so its cardinality is 1.

My answer

Q20. Quiz 2 Alenzi

Let $S = \{a, b\}$. The power set $P(S)$ is

(A) $\{\{a\}, \{b\}\}$

(B) $\{\emptyset, a, b, \{a, b\}\}$

(C) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

(D) $\{\{a\}, \{b\}, \{a, b\}\}$

Answer: (C). The power set lists all subsets of S, including $\emptyset$ and S itself; its elements are sets, not a and b.

My answer

Q21.

Quiz 2 Alenzi

Let $S = \{a, b\}$. What is $|P(P(S))|$ (the cardinality of the set $P(P(S))$)?

(A) 4

(B) 8

(C) 32

(D) 16

Answer: (D). $|S| = 2$, so $|P(S)| = 2^2 = 4$ and $|P(P(S))| = 2^{|P(S)|} = 2^4 = 16$.

My answer

Q22.

[new]

If $|A| = 5$, then $|P(A)| =$

(A) 10

(B) 25

(C) 32

(D) 120

Answer: (C). $|P(A)| = 2^{|A|} = 2^5 = 32$.

My answer

Q23.

[new]

The power set of the empty set, $P(\emptyset)$, is

(A) $\emptyset$

(B) $\{\emptyset\}$

(C) $\{\emptyset, \{\emptyset\}\}$

(D) $\{\{\emptyset\}\}$

Answer: (B). The only subset of $\emptyset$ is $\emptyset$, so $P(\emptyset) = \{\emptyset\}$, which has $2^0 = 1$ element.

My answer

Q24. [new]

$P(\{\emptyset\}) =$

(A) $\{\emptyset\}$

(B) $\{\{\emptyset\}\}$

(C) $\{\emptyset, \{\{\emptyset\}\}\}$

(D) $\{\emptyset, \{\emptyset\}\}$

Answer: (D). The subsets of $\{\emptyset\}$ are $\emptyset$ and $\{\emptyset\}$ itself.

My answer

Q25. [new]

Let $A = \{1, 2, 3\}$. Which of the following is an element of $P(A)$?

(A) $\{1, 3\}$

(B) 2

(C) $\{1, 4\}$

(D) $\{\{1\}\}$

Answer: (A). Elements of $P(A)$ are subsets of A; $\{1, 3\} \subseteq A$. 2 is an element of A, not a subset of A; $\{1, 4\} \not\subseteq A$, and $\{1\} \notin A$.

My answer

Q26. [new]

Let $A = \{1, 2, 3\}$. Which statement is FALSE?

(A) $\emptyset \in P(A)$

(B) $A \in P(A)$

(C) $\{\{1\}, \{2\}\} \subseteq P(A)$

(D) $\{1\} \subseteq P(A)$

Answer: (D). $\{1\} \subseteq P(A)$ would need $1 \in P(A)$, but the elements of $P(A)$ are sets. (Note $\{1\} \in P(A)$ is true.)

My answer

Q27. [new]

How many elements of $P(\{a, b, c\})$ contain the element a?

(A) 3

(B) 4

(C) 6

(D) 8

Answer: (B). $\{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}$: one for each subset of $\{b, c\}$, so $2^2 = 4$.

My answer

Q28. [new]

Which statement is true for every finite set A?

(A) $A \in P(A)$

(B) $A \subseteq P(A)$

(C) $\emptyset \notin P(A)$

(D) $|P(A)| = |A|^2$

Answer: (A). Every set is a subset of itself, so A is an element of $P(A)$. (D) fails for $A = \emptyset$: $1 \neq 0$; (B) fails for $A = \{1\}$.

My answer

Q29. Quiz 1 Alenzi

Which of the following statements is FALSE?

(A) $2 \in A \cup B$ implies that if $2 \notin A$ then $2 \in B$.

(B) $\{2, 3\} \subseteq A$ implies that $2 \in A$ and $3 \in A$.

(C) $A \cap B \supseteq \{2, 3\}$ implies that $\{2, 3\} \subseteq A$ and $\{2, 3\} \subseteq B$.

(D) $\{2\} \in A$ and $\{3\} \in A$ implies that $\{2, 3\} \subseteq A$.

Answer: (D). If the singleton $\{2\} \in A$, this does not imply that the element $2 \in A$ (e.g. $A = \{\{2\}, \{3\}\}$). The others follow from the definitions of union, intersection and subset.

My answer

Q30. [new]

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$. Then $A \cap B =$

(A) $\{1, 2, 3\}$

(B) $\{1, 2, 3, 4, 5, 6, 7\}$

(C) $\{6, 7\}$

(D) $\{4, 5\}$

Answer: (D). Only 4 and 5 are elements of both sets.

My answer

Q31. [new]

Let $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$ and $C = \{3, 4, 5, 6\}$. Then $A \cap B \cap C =$

(A) $\{4\}$

(B) $\{2, 4\}$

(C) $\{4, 6\}$

(D) $\emptyset$

Answer: (A). 4 is the only element common to all three sets.

My answer

Q32. [new]

Let $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$ and $C = \{3, 4, 5, 6\}$. Then $(A \cup B) \cap C =$

(A) $\{4\}$

(B) $\{3, 4, 6\}$

(C) $\{2, 3, 4, 6, 8\}$

(D) $\{3, 4, 5, 6\}$

Answer: (B). $A \cup B = \{1, 2, 3, 4, 6, 8\}$; intersecting with C gives $\{3, 4, 6\}$.

My answer

Q33. [new]

Let $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$ and $C = \{3, 4, 5, 6\}$. Then $A \cup (B \cap C) =$

(A) $\{3, 4, 6\}$

(B) $\{3, 4\}$

(C) $\{2, 3, 4, 6, 8\}$

(D) $\{1, 2, 3, 4, 5, 6, 8\}$

Answer: (C). $B \cap C = \{3, 4\}$, and $A \cup \{3, 4\} = \{2, 3, 4, 6, 8\}$. Compare with the previous question: parentheses matter.

My answer

Q34. [new]

Let $A = \{x \in \mathbb{Z} : x \text{ is a multiple of } 4\}$ and $B = \{x \in \mathbb{Z} : x \text{ is a multiple of } 6\}$. Then $A \cap B$ is

(A) the set of all multiples of 12

(B) the set of all multiples of 2

(C) the set of all multiples of 24

(D) $\{12\}$

Answer: (A). An integer is a multiple of both 4 and 6 exactly when it is a multiple of 12 (e.g. $\ldots, -12, 0, 12, 24, \ldots$).

My answer

Q35. [new]

Let $A = \{x \in \mathbb{R} : 0 \leq x < 3\}$ and $B = \{x \in \mathbb{R} : 2 < x \leq 5\}$. Then $A \cap B =$

(A) $\{x \in \mathbb{R} : 2 < x < 3\}$

(B) $\{x \in \mathbb{R} : 2 \leq x \leq 3\}$

(C) $\{x \in \mathbb{R} : 0 \leq x \leq 5\}$

(D) $\emptyset$

Answer: (A). x must satisfy $0 \leq x < 3$ and $2 < x \leq 5$, i.e. $2 < x < 3$.

My answer

Q36. Quiz 2 Alenzi

Let $A_n = \{n, n+1, n+2, \ldots\} \subset \mathbb{Z}^+$. What is $\bigcap_{n=1}^{\infty} A_n$?

(A) $\mathbb{Z}^+$

(B) $\emptyset$

(C) $\{1\}$

(D) $\{0\}$

Answer: (B). Suppose $m \in \bigcap_{n=1}^{\infty} A_n$. Then $m \in A_{m+1} = \{m+1, m+2, \ldots\}$, a contradiction. So the intersection is empty.

My answer

Q37. HW 3 (Rosen 2.2 #50b)

Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i, $A_i = \{0, i\}$.

(A) union $= \mathbb{Z}^+$, intersection $= \{0\}$

(B) union $= \mathbb{N}$, intersection $= \{0\}$

(C) union $= \mathbb{N}$, intersection $= \emptyset$

(D) union $= \mathbb{Z}$, intersection $= \{0\}$

Answer: (B). Every A_i contains 0, and every positive integer i appears in A_i, so the union is $\{0, 1, 2, \ldots\} = \mathbb{N}$. $A_1 \cap A_2 = \{0\}$, so the intersection is $\{0\}$.

My answer

Q38.

HW 3 (Rosen 2.2 #50c)

Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i, $A_i = (0, i)$, that is, the set of real numbers x with $0 < x < i$.

(A) union $= \{x \in \mathbb{R} : 0 < x < 1\}$, intersection $= \mathbb{R}^+$

(B) union $= \mathbb{R}^+$, intersection $= \emptyset$

(C) union $= \{x \in \mathbb{R} : x > 0\}$, intersection $= \{x \in \mathbb{R} : 0 < x < 1\}$

(D) union $= \mathbb{R}$, intersection $= \{1\}$

Answer: (C). The sets are nested, $A_1 \subseteq A_2 \subseteq \cdots$, so the intersection is $A_1 = (0, 1)$. Any $x > 0$ lies in A_i for $i > x$, so the union is $\mathbb{R}^+$.

My answer

Q39.

HW 3 (Rosen 2.2 #50d)

Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i, $A_i = (i, \infty)$, that is, the set of real numbers x with $x > i$.

(A) union $= \mathbb{R}^+$, intersection $= \emptyset$

(B) union $= \{x \in \mathbb{R} : x > 1\}$, intersection $= \{x \in \mathbb{R} : x > 1\}$

(C) union $= \{x \in \mathbb{R} : x \geq 1\}$, intersection $= \emptyset$

(D) union $= \{x \in \mathbb{R} : x > 1\}$, intersection $= \emptyset$

Answer: (D). $A_1 \supseteq A_2 \supseteq \cdots$, so the union is $A_1 = (1, \infty)$. No real x is larger than every positive integer i, so the intersection is $\emptyset$.

My answer

Q40.

[new]

For each positive integer i, let $A_i = \{x \in \mathbb{Z} : -i \leq x \leq i\}$. Then $\bigcap_{i=1}^{5} A_i =$

(A) $\{-5, -4, \ldots, 4, 5\}$

(B) $\{0\}$

(C) $\{-1, 0, 1\}$

(D) $\emptyset$

Answer: (C). The sets are nested, $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_5$, so the intersection is the smallest set $A_1 = \{-1, 0, 1\}$.

My answer

Q41. [new]

For each positive integer i, let $A_i = \{i, 2i\}$. Then $\bigcup_{i=1}^{3} A_i =$

(A) $\{1, 2, 3, 4, 6\}$

(B) $\{1, 2, 3, 4, 5, 6\}$

(C) $\{2, 4, 6\}$

(D) $\emptyset$

Answer: (A). $\{1, 2\} \cup \{2, 4\} \cup \{3, 6\} = \{1, 2, 3, 4, 6\}$.

My answer

Q42. [new]

For each positive integer i, let A_i be the set of all positive integer multiples of i. Then $\bigcap_{i=1}^{3} A_i$ is

(A) the set of all positive multiples of 3

(B) $\{6\}$

(C) the set of all positive multiples of 6

(D) $\mathbb{Z}^+$

Answer: (C). A positive integer is a multiple of 1, 2 and 3 exactly when it is a multiple of 6.

My answer

Q43. [new]

The universal set is the set of words in the Oxford English Dictionary. For each positive integer k, W_k is the set of words with exactly k letters. The set $\bigcup_{k=1}^{5} W_k$ is

(A) the set of all words with exactly 5 letters

(B) the set of all words with 5 or fewer letters

(C) the set of all words with at least 5 letters

(D) the empty set

Answer: (B). A word is in the union if it has k letters for at least one $k \in \{1, \ldots, 5\}$. (The intersection of these sets would be empty.)

My answer

Q44. [new]

If $A \subseteq B$, then $A \cup B =$

(A) A

(B) $\emptyset$

(C) $A \cap B$

(D) B

Answer: (D). Every element of A is already in B, so the union adds nothing new to B.

My answer

Q45. [new]

Let $A = \{1, 2\}$ and $B = \{2, 3\}$. Then $P(A) \cap P(B) =$

(A) $\{\{2\}\}$

(B) $\{\emptyset, \{2\}\}$

(C) $\{2\}$

(D) $\{\emptyset, \{1\}, \{2\}, \{3\}\}$

Answer: (B). The common subsets of A and B are $\emptyset$ and $\{2\}$ (these are exactly the subsets of $A \cap B = \{2\}$).

My answer

Q46. [new]

Which equation is true for all sets A and B?

(A) $P(A \cup B) = P(A) \cup P(B)$

(B) $P(A) \cup P(B) = P(A) \cap P(B)$

(C) $P(A \cap B) = P(A) \cap P(B)$

(D) $P(A) - P(B) = P(A - B)$

Answer: (C). $X \subseteq A \cap B$ iff $X \subseteq A$ and $X \subseteq B$. (A) fails for $A = \{1\}, B = \{2\}$: $\{1, 2\} \in P(A \cup B)$ only.

My answer

Q47. [new]

Let $A = \{a, b, c, e, f\}$ and $B = \{d, e, f, g\}$. Then $A - B =$

(A) $\{d, g\}$

(B) $\{a, b, c\}$

(C) $\{e, f\}$

(D) $\{a, b, c, d, g\}$

Answer: (B). Remove from A the elements also in B (e and f).

My answer

Q48. [new]

Let $A = \{a, b, c, e, f\}$ and $B = \{d, e, f, g\}$. Then $A \oplus B =$

(A) $\{a, b, c, d, g\}$

(B) $\{e, f\}$

(C) $\{a, b, c\}$

(D) $\{a, b, c, d, e, f, g\}$

Answer: (A). The symmetric difference contains the elements in exactly one of the two sets: $(A \cup B) - (A \cap B)$.

My answer

Q49. [new]

Let $A = \{10, 19, d, f\}$ and $B = \{5, 19, f\}$. Then $B - A =$

(A) $\{10, d\}$

(B) $\{19, f\}$

(C) $\{5\}$

(D) $\{5, 10, d\}$

Answer: (C). 19 and f are in A, so only 5 remains.

My answer

Q50. [new]

Let $A = \{10, 19, d, f\}$ and $B = \{5, 19, f\}$. Then $A \oplus B =$

(A) $\{19, f\}$

(B) $\{10, d\}$

(C) $\{5, 10, 19, d, f\}$

(D) $\{5, 10, d\}$

Answer: (D). $A \oplus B = (A - B) \cup (B - A) = \{10, d\} \cup \{5\}$.

My answer

Q51. [new]

Let the universal set be $U = \mathbb{Z}$ and let $A = \{x \in \mathbb{Z} : x \text{ is odd}\}$. Then $\overline{A}$ is

(A) the set of all even integers

(B) $\emptyset$

(C) the set of all positive even integers

(D) $\mathbb{Z}$

Answer: (A). $\overline{A} = U - A$: the integers that are not odd, i.e. the even integers (including 0 and negatives).

My answer

Q52. [new]

Which statement is true for all sets A, B and C?

(A) $A - B = B - A$

(B) $A \oplus B = B \oplus A$

(C) $(A - B) - C = A - (B - C)$

(D) $A - \emptyset = \emptyset$

Answer: (B). Symmetric difference is commutative. Set difference is neither commutative nor associative, and $A - \emptyset = A$.

My answer

Q53. [new]

Let $A = \{1, 2, 3\}$, $B = \{2, 3\}$ and $C = \{3\}$. Which pair gives $(A - B) - C$ and $A - (B - C)$, in that order?

(A) $\{1\}$ and $\{1\}$

(B) $\{1, 3\}$ and $\{1\}$

(C) $\{1\}$ and $\{1, 3\}$

(D) $\emptyset$ and $\{1, 3\}$

Answer: (C). $A - B = \{1\}$, so $(A - B) - C = \{1\}$. $B - C = \{2\}$, so $A - (B - C) = \{1, 3\}$. Set difference is not associative.

My answer

Q54. [new]

Let $U = \{1, 2, \ldots, 8\}$, $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Then $\overline{A \cup B} =$

(A) $\{3, 4\}$

(B) $\{1, 2, 5, 6, 7, 8\}$

(C) $\{1, 2, 5, 6\}$

(D) $\{7, 8\}$

Answer: (D). $A \cup B = \{1, \ldots, 6\}$, so its complement in U is $\{7, 8\}$.

My answer

Q55. [new]

Let $U = \{1, 2, \ldots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{2, 4, 6, 8\}$. Then $\overline{(A \oplus B)} \cap C =$

(A) $\{4, 8\}$

(B) $\{2, 6\}$

(C) $\{3, 4, 7, 8\}$

(D) $\{2, 4, 6, 8\}$

Answer: (A). $A \oplus B = \{1, 2, 5, 6\}$, $\overline{A \oplus B} = \{3, 4, 7, 8\}$, and intersecting with C gives $\{4, 8\}$.

My answer

Q56. [new]

Let A, B and C be finite sets. Which elements belong to $A \oplus B \oplus C$?

(A) The elements in exactly one of the three sets.

(B) The elements in exactly two of the three sets.

(C) The elements in all three sets.

(D) The elements in exactly one of the sets or in all three of the sets.

Answer: (D). An element is in $A \oplus B \oplus C$ iff it lies in an odd number of the sets (one or three).

My answer

Q57. [new]

$\{1, 2, 3\} \oplus \{2, 3, 4\} \oplus \{3, 4, 5\} =$

(A) $\{1, 5\}$

(B) $\{1, 3, 5\}$

(C) $\{3\}$

(D) $\{1, 2, 4, 5\}$

Answer: (B). $\{1, 2, 3\} \oplus \{2, 3, 4\} = \{1, 4\}$, then $\{1, 4\} \oplus \{3, 4, 5\} = \{1, 3, 5\}$. (3 is in all three sets.)

My answer

Q58. [new]

Let A be a subset of the universal set U. Which statement is FALSE?

(A) $A \oplus A = \emptyset$

(B) $A \oplus \emptyset = A$

(C) $A \oplus U = U$

(D) $A \oplus \overline{A} = U$

Answer: (C). $A \oplus U = U - A = \overline{A}$, which equals U only when $A = \emptyset$.

My answer

Q59. [new]

In a Venn diagram with three overlapping circles A, B, C, the shaded region is the part of circle A that does not overlap B or C. Which set does the shaded region represent?

(A) $A \cap \overline{B \cap C}$

(B) $A - (B \cup C)$

(C) $A \oplus B \oplus C$

(D) $(A - B) \cup C$

Answer: (B). The elements of A that are in neither B nor C. (A) would also include the parts of A inside exactly one of B, C.

My answer

Q60. Quiz 2 Alenzi

Which choice gives an example of sets A, B, C such that $B \neq C$ and $B - A = C - A$?

(A) $A = \{1\}$, $B = \{1, 2\}$, $C = \{1, 3\}$

(B) $A = \{1, 2\}$, $B = \{3\}$, $C = \{3, 4\}$

(C) $A = \{1, 2\}$, $B = \{1, 2, 9\}$, $C = \{2, 9\}$

(D) $A = \emptyset$, $B = \{1\}$, $C = \{2\}$

Answer: (C). $B \neq C$ since $1 \in B$, $1 \notin C$, and $B - A = \{9\} = C - A$. The idea is to let B and C differ only on elements inside A.

My answer

Q61.

HW 3 (Rosen 2.2 #14)

Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$, and $A \cap B = \{3, 6, 9\}$.

(A) $A = \{1, 5, 7, 8\}$, $B = \{2, 10\}$

(B) $A = \{1, 3, 5, 6, 7, 8, 9\}$, $B = \{2, 10\}$

(C) $A = \{1, 2, 5, 7, 8, 10\}$, $B = \{3, 6, 9\}$

(D) $A = \{1, 3, 5, 6, 7, 8, 9\}$, $B = \{2, 3, 6, 9, 10\}$

Answer: (D). $A = (A - B) \cup (A \cap B)$ and $B = (B - A) \cup (A \cap B)$.

My answer

Q62.

Quiz 1 Alenzi

Select the definitions for sets A and B that show the set equation $(B - A) \cup A = B$ is not a set identity.

(A) $A = \{1\}$ and $B = \{1\}$

(B) $A = \{2, 4, 5\}$ and $B = \{1, 2, 3, 4, 5\}$

(C) $A = \{1\}$ and $B = \{1, 2\}$

(D) $A = \{1, 2\}$ and $B = \{2, 3\}$

Answer: (D). $B - A = \{3\}$ and $(B - A) \cup A = \{1, 2, 3\} \neq B$. In (A)–(C), $A \subseteq B$, and then the equation holds.

My answer

Q63.

[new]

Which set identity justifies $A \cup (A \cap B) = A$?

(A) Distributive law

(B) Idempotent law

(C) Absorption law

(D) Domination law

Answer: (C). The absorption laws are $A \cup (A \cap B) = A$ and $A \cap (A \cup B) = A$.

My answer

Q64. [new]

Which set identity justifies $\overline{A \cup B} = \overline{A} \cap \overline{B}$?

(A) De Morgan's law

(B) Complement law

(C) Double complement law

(D) Distributive law

Answer: (A). De Morgan's laws: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

My answer

Q65. [new]

Which set identity justifies $A \cup \overline{A} = U$?

(A) Identity law

(B) Domination law

(C) Idempotent law

(D) Complement law

Answer: (D). Complement laws: $A \cup \overline{A} = U$, $A \cap \overline{A} = \emptyset$ (and $\overline{U} = \emptyset$, $\overline{\emptyset} = U$).

My answer

Q66. [new]

Which identity is the double complement law?

(A) $\overline{\overline{A}} = A$

(B) $A \cap A = A$

(C) $\overline{U} = \emptyset$

(D) $A \cup \emptyset = A$

Answer: (A). (B) is idempotent, (C) is a complement law, (D) is an identity law.

My answer

Q67. [new]

Using the set identities, $\overline{A \cup \overline{B}}$ is equal to

(A) $\overline{A} \cup B$

(B) $A \cap \overline{B}$

(C) $\overline{A} \cap B$

(D) $\overline{A} \cap \overline{B}$

Answer: (C). De Morgan: $\overline{A} \cap \overline{\overline{B}}$; double complement: $\overline{A} \cap B$.

My answer

Q68. [new]

Using the set identities, $(A \cap B) \cup (A \cap \overline{B})$ simplifies to

(A) A

(B) B

(C) U

(D) $\emptyset$

Answer: (A). Distributive: $A \cap (B \cup \overline{B})$; complement: $A \cap U$; identity: A.

My answer

Q69. [new]

Using the set identities, $A \cap (\overline{A} \cup B)$ simplifies to

(A) A

(B) $A \cup B$

(C) $\emptyset$

(D) $A \cap B$

Answer: (D). Distributive: $(A \cap \overline{A}) \cup (A \cap B) = \emptyset \cup (A \cap B) = A \cap B$ (complement, then identity law).

My answer

Q70. [new]

In proving $\overline{A \cap B} = \overline{A} \cup \overline{B}$ with logic, the step from $\neg(x \in A \wedge x \in B)$ to $\neg(x \in A) \vee \neg(x \in B)$ is justified by

(A) the definition of set complement

(B) De Morgan's law for propositions

(C) the definition of set union

(D) the distributive law for propositions

Answer: (B). $\neg(p \wedge q) \equiv \neg p \vee \neg q$ with p: $x \in A$, q: $x \in B$.

My answer

Q71. [new]

Which of the following set equations is NOT a set identity?

(A) $A - B = A \cap \overline{B}$

(B) $(A \cup B) - B = A$

(C) $A \cup (A \cap B) = A$

(D) $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Answer: (B). With $A = B = \{1\}$: $(A \cup B) - B = \emptyset \neq A$. In general $(A \cup B) - B = A - B$.

My answer

Q72. [new]

Select the definitions of A, B, C that show the equation $A \cup (B \cap C) = (A \cup B) \cap C$ is not a set identity.

(A) $A = \{1\}$, $B = \{1\}$, $C = \{1\}$

(B) $A = \emptyset$, $B = \{1\}$, $C = \{2\}$

(C) $A = \{1\}$, $B = \emptyset$, $C = \emptyset$

(D) $A = \emptyset$, $B = \{1\}$, $C = \{1\}$

Answer: (C). Left side: $\{1\} \cup \emptyset = \{1\}$. Right side: $\{1\} \cap \emptyset = \emptyset$. In (A), (B), (D) both sides are equal.

My answer

Q73. [new]

Using the set subtraction law $A - B = A \cap \overline{B}$ and the other identities, $A - (A - B)$ is equal to

(A) $\emptyset$

(B) $A \cap B$

(C) B

(D) $A - B$

Answer: (B). $A \cap \overline{A \cap \overline{B}} = A \cap (\overline{A} \cup B) = (A \cap \overline{A}) \cup (A \cap B) = A \cap B$.

My answer

Q74. [new]

When set identities are derived from the laws of logic, U corresponds to T and $\emptyset$ to F. Which logical equivalence corresponds to the set identity $A \cap U = A$?

(A) $p \vee T \equiv T$

(B) $p \wedge F \equiv F$

(C) $p \wedge T \equiv p$

(D) $p \vee F \equiv p$

Answer: (C). Intersection corresponds to $\wedge$ and U to T; both are identity laws.

My answer

Q75. [new]

Using the set identities, $\overline{A \cup B} \cup (\overline{A} \cap B)$ simplifies to

(A) $\overline{B}$

(B) U

(C) $\overline{A} \cap \overline{B}$

(D) $\overline{A}$

Answer: (D). De Morgan: $(\overline{A} \cap \overline{B}) \cup (\overline{A} \cap B) = \overline{A} \cap (\overline{B} \cup B) = \overline{A} \cap U = \overline{A}$.

My answer

Q76. [new]

Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then $A \times B =$

(A) $\{(1, a), (2, b), (2, c)\}$

(B) $\{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$

(C) $\{1, 2, a, b, c\}$

(D) $\{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

Answer: (D). All ordered pairs with first entry from A and second entry from B; $|A \times B| = 2 \cdot 3 = 6$. (B) is $B \times A$.

My answer

Q77. [new]

If $|A| = 4$ and $|B| = 3$, then $|A \times B| =$

(A) 7

(B) 12

(C) 64

(D) 81

Answer: (B). $|A \times B| = |A| \cdot |B| = 12$.

My answer

Q78. [new]

Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Which statement is true?

(A) $(a, 1) \in B \times A$

(B) $(a, 1) \in A \times B$

(C) $A \times B = B \times A$

(D) $|B \times A| = 5$

Answer: (A). The first entry of a pair in $B \times A$ comes from B. $A \times B \neq B \times A$ since order matters, and $|B \times A| = 6$.

My answer

Q79. [new]

If $(2x - 1,\ y + 3) = (5, 7)$, then

(A) $x = 2,\ y = 4$

(B) $x = 3,\ y = 10$

(C) $x = 3,\ y = 4$

(D) $x = 4,\ y = 3$

Answer: (C). Ordered pairs are equal iff their first entries and second entries are equal: $2x - 1 = 5$ and $y + 3 = 7$.

My answer

Q80. [new]

For any set A, $A \times \emptyset =$

(A) A

(B) $\emptyset$

(C) $\{\emptyset\}$

(D) $\{(a, \emptyset) : a \in A\}$

Answer: (B). A pair $(a, b) \in A \times \emptyset$ would need $b \in \emptyset$, which is impossible.

My answer

Q81. [new]

Which of the following is an element of $\mathbb{Z} \times \mathbb{N}$?

(A) $(0, -3)$

(B) $(\frac{1}{2}, 1)$

(C) $(-3, 0)$

(D) $(2, -1)$

Answer: (C). The first entry must be an integer and the second a natural number (a nonnegative integer); $-3 \in \mathbb{Z}$ and $0 \in \mathbb{N}$.

My answer

Q82. [new]

Let $A = \{a\}$, $B = \{1, 2\}$ and $C = \{x, y\}$. Which of the following is an element of $A \times B \times C$?

(A) $(2, a, x)$

(B) $(a, 2, x)$

(C) $(a, x, 2)$

(D) $(a, 2)$

Answer: (B). Elements are ordered triples with entries from A, B, C in that order. (D) is not a triple.

My answer

Q83. [new]

How many elements does $\{0, 1\}^3$ have?

(A) 3

(B) 6

(C) 8

(D) 9

Answer: (C). $|\{0, 1\}^3| = 2^3 = 8$: the binary strings $000, 001, \ldots, 111$.

My answer

Q84. [new]

Express $\{x \in \{0, 1\}^3 : x \text{ has exactly two 1's}\}$ in roster notation (elements as strings).

(A) $\{011, 101, 110\}$

(B) $\{11, 011, 110\}$

(C) $\{011, 101, 110, 111\}$

(D) $\{001, 010, 100\}$

Answer: (A). The 3-bit strings with exactly two 1's.

My answer

Q85. [new]

Define the strings $s = 011$ and $t = 10$. What is the concatenation st?

(A) 10011

(B) 01110

(C) 0111

(D) 01101

Answer: (B). st is s followed by t: $011\,10$. Its length is $3 + 2 = 5$.

My answer

Q86. [new]

Let λ denote the empty string. Which statement is true?

(A) $\{0,1\}^0 = \emptyset$

(B) $\{0,1\}^0 = \{0,1\}$

(C) $|\{0,1\}^0| = 0$

(D) $\{0,1\}^0 = \{\lambda\}$

Answer: (D). The only binary string of length 0 is the empty string λ, so $|\{0,1\}^0| = 1$.

My answer

Q87. [new]

Which equation is true for all sets A, B and C?

(A) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(B) $A \times B = B \times A$

(C) $A \times (B \cup C) = (A \times B) \cup C$

(D) $(A \times B) \cap (B \times A) = \emptyset$

Answer: (A). $(x, y) \in A \times (B \cup C)$ iff $x \in A$ and ($y \in B$ or $y \in C$) iff $(x, y) \in A \times B$ or $(x, y) \in A \times C$.

My answer

Q88. [new]

Let A and B be sets with $A \times B = \emptyset$. What can be concluded?

(A) $A = \emptyset$ and $B = \emptyset$

(B) $A = B$

(C) $A = \emptyset$ or $B = \emptyset$

(D) $A \cap B = \emptyset$

Answer: (C). If both were nonempty, picking $a \in A$, $b \in B$ gives $(a, b) \in A \times B$. Only one of them needs to be empty.

My answer

Q89. [new]

If $|A| = 2$ and $|B| = 3$, what is $|P(A \times B)|$?

(A) 6

(B) 12

(C) 32

(D) 64

Answer: (D). $|A \times B| = 6$, so $|P(A \times B)| = 2^6 = 64$.

My answer

Q90. [new]

Let $A = \{1, 2\}$ and $B = \{2, 3\}$. Then $(A \times B) \cap (B \times A) =$

(A) $\emptyset$

(B) $\{(2, 2)\}$

(C) $\{2\}$

(D) $\{(1, 2), (2, 2), (2, 3)\}$

Answer: (B). A pair in both must have both entries in $A \cap B = \{2\}$.

My answer

Part B: Written problems

Q91.

HW 3 (Rosen 2.1 #22)

Can you conclude that $A = B$ if A and B are two sets with the same power set? Justify your answer.

Answer:

Yes. Suppose $P(A) = P(B)$.

Since $A \subseteq A$, we have $A \in P(A) = P(B)$, so A is a subset of B: $A \subseteq B$.

Similarly, $B \subseteq B$ gives $B \in P(B) = P(A)$, so $B \subseteq A$.

Two sets are equal iff each is a subset of the other, so $A = B$. ■

My answer

Q92.

HW 3 (Rosen 2.2 #30c)

Can you conclude that $A = B$ if A, B and C are sets such that $A \cup C = B \cup C$ and $A \cap C = B \cap C$? Justify your answer.

Answer:

Yes. We show $A \subseteq B$; the proof of $B \subseteq A$ is the same with A and B swapped.

Let $x \in A$. There are two cases.

- Case 1: $x \in C$. Then $x \in A \cap C = B \cap C$, so $x \in B$.
- Case 2: $x \notin C$. Since $x \in A$, $x \in A \cup C = B \cup C$. So $x \in B$ or $x \in C$. Because $x \notin C$, $x \in B$.

In both cases $x \in B$, so $A \subseteq B$. By symmetry $B \subseteq A$, hence $A = B$. ■

(Neither condition alone is enough. For example, $A = \{1\}$, $B = \emptyset$, $C = \{1\}$ gives $A \cup C = B \cup C$ but $A \neq B$. $A = \{1\}$, $B = \{2\}$, $C = \emptyset$ gives $A \cap C = B \cap C$ but $A \neq B$.)

My answer

Q93. Quiz 2 Alenzi

Let A, B, and C be sets. Show that

$$A \times (B \cap C) = (A \times B) \cap (A \times C).$$

Answer:

We show that a pair (x, y) is in the left side iff it is in the right side. Every element of either side is an ordered pair.

$$\begin{aligned}
(x,y) \in A \times (B \cap C) &\iff x \in A \ \wedge\ y \in B \cap C && \text{(def. of Cartesian product)} \\
&\iff x \in A \ \wedge\ (y \in B \wedge y \in C) && \text{(def. of intersection)} \\
&\iff (x \in A \wedge y \in B) \wedge (x \in A \wedge y \in C) && \text{(since } p \equiv p \wedge p\text{; assoc./comm.)} \\
&\iff (x,y) \in A \times B \ \wedge\ (x,y) \in A \times C && \text{(def. of Cartesian product)} \\
&\iff (x,y) \in (A \times B) \cap (A \times C) && \text{(def. of intersection)}
\end{aligned}$$

So each side is a subset of the other (mutual inclusion), and $A \times (B \cap C) = (A \times B) \cap (A \times C)$. ■

My answer

Q94. Quiz 3 Adel

If A, B and C are sets, then prove or disprove

$$\overline{A \times (B \cup C)} = A \times \overline{(B \cup C)}.$$

Answer:

False. The complement on the left is taken in the product universe $U \times U$. The complement $\overline{B \cup C}$ on the right is taken in U and then paired with A.

Counterexample: $U = \{a, b\}$, $A = \{a\}$, $B = \{b\}$, $C = \emptyset$.

- $B \cup C = \{b\}$, so $A \times (B \cup C) = \{(a, b)\}$. Then $\overline{A \times (B \cup C)} = (U \times U) - \{(a, b)\} = \{(a, a), (b, a), (b, b)\}$.
- $\overline{B \cup C} = U - \{b\} = \{a\}$, so $A \times \overline{(B \cup C)} = \{a\} \times \{a\} = \{(a, a)\}$.

The two sets are unequal (for example (b, b) is on the left only), so the identity fails. ■

My answer

Q95. Quiz 3 Adel

Use set builder notation and logical equivalences to establish that $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Answer:

$$
\begin{aligned}
\overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{(definition of complement)} \\
&= \{x \mid \neg(x \in A \cap B)\} && \text{(definition of } \notin) \\
&= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{(definition of intersection)} \\
&= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{(De Morgan's law, propositional)} \\
&= \{x \mid x \notin A \vee x \notin B\} && \text{(definition of } \notin) \\
&= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} && \text{(definition of complement)} \\
&= \{x \mid x \in \overline{A} \cup \overline{B}\} && \text{(definition of union)} \\
&= \overline{A} \cup \overline{B} && \text{(meaning of set builder notation)} \quad \blacksquare
\end{aligned}
$$

My answer

Q96. **[new]**

Use the set identities from the table to prove the identity below. Label each step with the identity used.

$$\overline{A \cup (B \cap C)} = (\overline{A} \cap \overline{B}) \cup (\overline{A} \cap \overline{C}).$$

Answer:

$$\begin{aligned} \overline{A \cup (B \cap C)} &= \overline{A} \cap \overline{B \cap C} && \text{(De Morgan's law)} \\ &= \overline{A} \cap (\overline{B} \cup \overline{C}) && \text{(De Morgan's law)} \\ &= (\overline{A} \cap \overline{B}) \cup (\overline{A} \cap \overline{C}) && \text{(Distributive law)} \end{aligned}$$ ■

My answer

Q97. **[new]**

The set subtraction law says that $X - Y = X \cap \overline{Y}$. Use it and the set identities to prove

$$A - (B \cup C) = (A - B) \cap (A - C).$$

Label each step.

Answer:

$$\begin{aligned}
A - (B \cup C) &= A \cap \overline{B \cup C} && \text{(Set subtraction law)} \\
&= A \cap (\overline{B} \cap \overline{C}) && \text{(De Morgan's law)} \\
&= (A \cap A) \cap (\overline{B} \cap \overline{C}) && \text{(Idempotent law)} \\
&= (A \cap \overline{B}) \cap (A \cap \overline{C}) && \text{(Associative and commutative laws)} \\
&= (A - B) \cap (A - C) && \text{(Set subtraction law)} \quad \blacksquare
\end{aligned}$$

My answer

Q98. **[new]**

Show that each set equation below is not a set identity by giving sets that make it false. Compute both sides.

(a) $A - (B - C) = (A - B) - C$

(b) $A \cup (B \cap C) = (A \cup B) \cap C$

(c) $(A \cup B) - B = A$

Answer:

(a) Take $A = B = C = \{1\}$. Then $B - C = \emptyset$, so $A - (B - C) = \{1\}$. But $A - B = \emptyset$, so $(A - B) - C = \emptyset$. Since $\{1\} \neq \emptyset$, the equation is not an identity.

(b) Take $A = \{1\}$, $B = C = \emptyset$. Then $A \cup (B \cap C) = \{1\} \cup \emptyset = \{1\}$, but $(A \cup B) \cap C = \{1\} \cap \emptyset = \emptyset$.

(c) Take $A = B = \{1\}$. Then $(A \cup B) - B = \{1\} - \{1\} = \emptyset \neq \{1\} = A$.

(In general $(A \cup B) - B = A - B$, which equals A only when $A \cap B = \emptyset$.) $\blacksquare$

My answer

Q99. **[new]**

Let $A = \{1, 2\}$ and $B = \{2, 3\}$.

(a) Write $P(A)$, $P(B)$ and $P(A) \cap P(B)$ in roster notation, and compare $P(A) \cap P(B)$ with $P(A \cap B)$.

(b) Find $|P(A) \cup P(B)|$ and $|P(A \cup B)|$. Is $P(A) \cup P(B) = P(A \cup B)$?

(c) Write $A \times B$, $B \times A$ and $(A \times B) \cap (B \times A)$ in roster notation.

(d) Find $|P(A \times B)|$.

Answer:

(a) $P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ and $P(B) = \{\emptyset, \{2\}, \{3\}, \{2, 3\}\}$, so $P(A) \cap P(B) = \{\emptyset, \{2\}\}$. Since $A \cap B = \{2\}$, $P(A \cap B) = \{\emptyset, \{2\}\}$: the two sets are equal.

(b) $P(A) \cup P(B) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}\}$ has 6 elements. $A \cup B = \{1, 2, 3\}$, so $|P(A \cup B)| = 2^3 = 8$. They are not equal: $\{1, 3\}$ and $\{1, 2, 3\}$ are in $P(A \cup B)$ but not in $P(A) \cup P(B)$.

(c) $A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$, $B \times A = \{(2, 1), (2, 2), (3, 1), (3, 2)\}$, and $(A \times B) \cap (B \times A) = \{(2, 2)\}$.

(d) $|A \times B| = 2 \cdot 2 = 4$, so $|P(A \times B)| = 2^4 = 16$.

My answer

Q100. [new]

For each positive integer i, let $A_i = \{x \in \mathbb{R} : -\frac{1}{i} < x < \frac{1}{i}\}$. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$, and justify your answers.

Answer:

Union. Since $\frac{1}{i+1} < \frac{1}{i}$, the sets are nested: $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$. Every $A_i \subseteq A_1$, so $\bigcup_{i=1}^{\infty} A_i = A_1 = \{x \in \mathbb{R} : -1 < x < 1\}$.

Intersection. We claim $\bigcap_{i=1}^{\infty} A_i = \{0\}$.

- $0 \in A_i$ for every i, because $-\frac{1}{i} < 0 < \frac{1}{i}$. So $\{0\} \subseteq \bigcap A_i$.
- Let $x \neq 0$. Choose a positive integer $i > \frac{1}{|x|}$. Then $\frac{1}{i} < |x|$, so $x \notin A_i$, and hence $x \notin \bigcap A_i$.

Therefore $\bigcap_{i=1}^{\infty} A_i = \{0\}$. ■

My answer